SUPPLY CHAIN ANALYTICS

End-to-End Data Analyst Portfolio Project

DataCo Global Supply Chain Dataset

180,519 Records | 53 Columns | 65,752 Unique Orders

Skills Demonstrated: Python · Pandas · SQL · EDA · Feature Engineering · Business Intelligence · Storytelling

PHASE 1 — PROJECT PLANNING

1.1 Recommended Project Title

"Operational Excellence in Global Supply Chain: A Data-Driven Analysis of Delivery Performance, Profitability, and Fraud Risk at DataCo"

1.2 Business Background

DataCo Global is a multinational supply chain and logistics company operating across five major global markets: LATAM, Europe, Pacific Asia, USCA, and Africa. The company manages end-to-end order fulfillment — from customer order placement through product shipping — spanning 118 products across 50+ categories, serving 20,652 unique customers over a 4-year period (2015-2018).

1.3 Industry Context

The global supply chain management market was valued at ~$19B and is expected to grow at 11% CAGR. Key industry challenges include late delivery (avg industry rate: 23%), inventory shrinkage, fraud, and margin pressure from e-commerce discounting. This dataset reflects real-world operational complexity that Data Analysts must navigate.

1.4 Supply Chain Concepts Involved

- Order Fulfillment Cycle: Order → Processing → Shipping → Delivery
- Delivery Performance: Scheduled vs. Actual shipping days
- Profit Margin Management: Discount-driven margin erosion
- Customer Segmentation: Consumer, Corporate, Home Office
- Geographic Market Analysis: Regional performance benchmarking
- Fraud Detection: Suspected_Fraud order status with negative margins
- ABC Inventory Analysis: Pareto classification of SKUs by revenue

1.5 Business Problems Visible in the Dataset

1. 57.3% late delivery rate — a critical operational crisis
2. 18.7% of orders produce negative profit — pricing/discount policy failure
3. 2.25% suspected fraud rate — $830K+ annual revenue leakage
4. 86.2% missing Order Zipcode — data governance failure
5. 100% null Product Description — catalog management failure
6. Same Day shipping has worst on-time performance — SLA inversion problem

1.6 Project Objectives

7. Quantify and profile late delivery performance across markets, regions, and shipping modes

8. Identify profit-destroying products and discount thresholds
9. Detect and profile fraudulent order patterns
10. Build a KPI framework management can track on a dashboard
11. Develop SQL-ready data model for BI tool integration
12. Generate actionable recommendations with estimated business value

1.7 KPIs Management Would Track

KPI	Current Value	Target / Benchmark
Total Revenue	$36.78M	—
Total Profit	$3.97M	—
Profit Margin %	10.8%	>15% (industry avg)
Late Delivery Rate	57.3%	<20% (industry target)
Avg Shipping Days	3.5 days	3.0 days
Fraud Rate	2.25%	<0.5%
Negative Profit Rate	18.7%	<5%
Unique Customers	20,652	Growth target +15% YoY

1.8 Project Success Criteria

- Deliver a cleaned dataset with documented transformation logic
- Generate 20+ EDA visualizations with business interpretation
- Build 40+ SQL queries (Levels 1-3) covering all analytical needs
- Produce 25 management-level business insights
- Create an executive report suitable for C-suite presentation
- Prepare comprehensive interview Q&A covering Python, SQL, and domain knowledge

PHASE 2 — DATASET UNDERSTANDING

2.1 Dataset Overview

Attribute	Value
Total Records	180,519 rows
Columns	53 original → 47 after cleaning
Unique Orders	65,752
Unique Customers	20,652
Date Range	January 2015 – February 2018
Markets	5 (LATAM, Europe, Pacific Asia, USCA, Africa)
Products	118 unique SKUs across 50 categories
File Size	~313 MB in memory

2.2 Column Classification & Analysis

Columns are classified into 7 categories. Key decisions are documented below.

Customer Information Columns

Column	Type	Business Meaning	Decision	Reason
Customer Id	int	Unique customer identifier	KEEP	Primary key for customer analysis
Customer Fname/Lname	str	Customer name	KEEP	Useful for customer lookup
Customer Email	str	Contact email	DROP	100% same value — PII placeholder
Customer Password	str	Account password	DROP	Security risk, zero analytical value
Customer Segment	str	Business segment (Consumer/Corp/HO)	KEEP	Critical segmentation variable
Customer City/State/Zipcode	str/int	Geographic location	KEEP	Geographic segmentation

Financial Information Columns (High Value)

Column	Type	Business Meaning	Decision	Reason
Sales	float	Revenue per order item	KEEP	Primary revenue KPI

Column	Type	Business Meaning	Decision	Reason
Order Profit Per Order	float	Net profit per order	KEEP	Core profitability metric
Benefit per order	float	Gross benefit	KEEP	Alternative profitability view
Order Item Discount Rate	float	Discount % applied	KEEP	Key driver of margin erosion
Order Item Profit Ratio	float	Profit as % of item price	KEEP	Normalized profitability

Useless/Dropped Columns

Column	Problem	Decision
Product Description	100% null — no data	DROP
Customer Email	Single value — anonymized	DROP
Customer Password	PII security risk, no value	DROP
Product Image	URL strings — no analytical value	DROP
Product Status	All records = 0, zero variance	DROP
Order Zipcode	86.2% missing — unreliable	DROP

PHASE 3 — DATA QUALITY ASSESSMENT REPORT

3.1 Summary

Issue	Affected Records	Impact	Decision
Product Description — 100% null	180,519 rows	HIGH — no product content	DROP column
Order Zipcode — 86.2% null	155,679 rows	HIGH — no geo depth	DROP column
Customer Email — 1 unique value	180,519 rows	HIGH — PII placeholder	DROP column
Customer Password — 1 unique value	180,519 rows	CRITICAL — security	DROP column
Negative Profit Orders	33,784 rows (18.7%)	HIGH — margin loss	FLAG & analyze
Customer Lname — 8 null	8 rows (<0.01%)	LOW	Fill 'Unknown'
Customer Zipcode — 3 null	3 rows (<0.01%)	LOW	Fill 0
Duplicate rows	0	NONE	No action needed
Product Status — 1 unique value	180,519 rows	MEDIUM — zero variance	DROP column

3.2 Data Quality Assessment: Why This Matters

A data quality assessment is not just a technical step — it is a business risk assessment. Poor data quality leads to:

- Wrong KPIs reported to management → wrong strategic decisions
- Biased ML models trained on missing/incorrect features
- Compliance failures (storing plaintext passwords is a GDPR/security violation)
- Missed business opportunities due to incomplete geographic data

PHASE 4 — DATA CLEANING (Python)

4.1 Complete Python Cleaning Code

The following code represents production-quality data cleaning with full documentation.

```
import pandas as pd import numpy as np  # ── STEP 1: LOAD ──────────────────────────────────────── df = pd.read_csv('DataCoSupplyChainDataset.csv', encoding='latin1') print(f"Shape: {df.shape}")  # (180519, 53)  # ── STEP 2: DROP USELESS / PII COLUMNS ──────────────────────── # WHY: Customer Email and Password are anonymized placeholders #      (1 unique value each). Product Description is 100% null. #      Product Image is just URLs. Product Status has zero variance. cols_to_drop = [     'Customer Email', # PII placeholder - 1 unique value     'Customer Password', # Security risk - 1 unique value 'Product Description',# 100% null     'Product Image',     # URL strings     'Product Status',    # Zero variance (all = 0)     'Order Zipcode',     # 86.2% missing - unreliable ] df.drop(columns=cols_to_drop, inplace=True)  # ── STEP 3: PARSE DATE COLUMNS ──────────────────────────── # WHY: Stored as strings (format: 'MM/DD/YYYY HH:MM') # Must convert to datetime for time-series analysis. df['order_date'] = pd.to_datetime( df['order date (DateOrders)'], format='%m/%d/%Y %H:%M') df['shipping_date'] = pd.to_datetime( df['shipping date (DateOrders)'], format='%m/%d/%Y %H:%M') df.drop(columns=['order date (DateOrders)',                  'shipping date (DateOrders)'], inplace=True)  # ── STEP 4: HANDLE MISSING VALUES ──────────────────────────── # WHY: 8 missing Customer Lname - fill with 'Unknown' to preserve rows #      Dropping rows with <0.01% nulls wastes 99.99% good data. df['Customer Lname'].fillna('Unknown', inplace=True) df['Customer Zipcode'].fillna(0, inplace=True)  # ── STEP 5: STANDARDIZE TEXT COLUMNS ──────────────────────── # WHY: Text inconsistencies (mixed case, whitespace) cause #      incorrect GROUP BY results in SQL and Python. text_cols = ['Delivery Status', 'Shipping Mode', 'Market',              'Customer Segment', 'Order Status'] for col in text_cols:     df[col] = df[col].str.strip().str.title() # ── STEP 6: RENAME COLUMNS (Python Friendly) ──────────────── # WHY: Column names with spaces, parentheses, and mixed case #      require df['Column Name'] syntax. Python-style names #      enable df.column_name attribute access — cleaner code. df.rename(columns={ 'Days for shipping (real)': 'actual_shipping_days',     'Order Profit Per Order': 'profit_per_order',     'Order Item Discount Rate': 'discount_rate',     # ... [all 40+ columns renamed] }, inplace=True)  print(f"Clean shape: {df.shape}")  # (180519, 47) df.to_csv('supply_chain_clean.csv', index=False)
```

PHASE 5 — FEATURE ENGINEERING

5.1 New Features Created

Feature	Formula	Business Meaning	Expected Insight
delivery_delay_days	actual_days - scheduled_days	How many days over/under schedule	Negative = early, 0 = on time, positive = late
is_late_delivery	delivery_delay > 0 → 1	Binary flag for late orders	Enables rate calculation and ML targeting
profit_margin_pct	profit / sales × 100	Net profitability as percentage	Identify loss-making vs high-margin orders
is_high_value	order_total >= 75th pct	Premium order flag (top 25%)	Target for VIP treatment and premium shipping
order_year/month/quarter	Extracted from order_date	Time decomposition for trend analysis	Q4 peak detection, YoY growth
discount_band	Binned discount_rate	Categorical discount tier	Correlate discount depth with profit erosion
shipping_speed	Binned actual_days	Express / Standard / Slow	Operational efficiency categorization
risk_category	Fraud status + late risk	High / Medium / Low risk	Priority queue for operations review
revenue_per_unit	sales / quantity	Average selling price per unit	Price consistency and premium detection

5.2 Why Feature Engineering Matters

Raw data rarely contains the exact variables that answer business questions. Feature engineering transforms available data into analytical signals. For example:

- delivery_delay_days converts two separate date columns into a single actionable metric that directly measures SLA compliance.
- discount_band converts a continuous variable into interpretable categories, enabling CASE WHEN logic in SQL dashboards.
- is_late_delivery creates a binary target variable suitable for classification models (future ML extension).
- risk_category combines multiple signals into a single operational priority indicator used by logistics teams.

PHASE 6 — EXPLORATORY DATA ANALYSIS

Chart 1: Total Sales by Market

Business Question	Which geographic market generates the most revenue?
Insight	LATAM leads with highest total sales ($9.5M+), followed closely by Europe. Africa is the smallest market at ~$2.1M, representing a significant growth opportunity.
Business Impact	Resource allocation: marketing and logistics investment should be proportional to market revenue potential.
Recommendation	Maintain LATAM and Europe as core markets while developing a strategic plan for Africa market expansion.



Chart 2: Total Profit by Market

Business Question	Which market is most profitable, and does it align with sales rank?
Insight	Europe generates the highest profit despite being second in sales — indicating superior operational efficiency. LATAM's profit does not match its revenue rank, suggesting higher costs or deeper discounting.
Business Impact	Profit-to-revenue ratio reveals operational quality differences between markets.
Recommendation	Study Europe's operating model as a benchmark. Investigate LATAM cost structure for optimization.



Chart 3: Late Delivery Rate by Shipping Mode

Business Question	Which shipping mode has the worst on-time delivery performance?
Insight	CRITICAL FINDING: Same Day shipping has the highest late delivery rate. This is the worst possible outcome — customers paying the highest premium are experiencing the worst service.
Business Impact	Severe brand damage and customer churn risk for premium customers. Potential for negative viral reviews.
Recommendation	Immediately audit Same Day fulfillment process. Either fix within 30 days or discontinue the service until resolved.



Chart 4: Monthly Sales Trend

Business Question	Does revenue show seasonality or a growth trend over time?
Insight	Sales show consistent Q4 peaks across all years, with relative stability at other times. The time series shows no dramatic downward trend, suggesting stable business operations.
Business Impact	Predictable seasonality enables proactive supply chain preparation.
Recommendation	Build inventory buffers 8-10 weeks before Q4 peak. Pre-negotiate carrier capacity in August.



Chart 5: Delivery Status Distribution

Business Question	How are deliveries distributed across performance categories?
Insight	Late delivery represents 54.8% of all orders — the dominant category. Only 17.8% shipped on time, and 23% actually arrived early (Advance shipping).
Business Impact	More than half of all customers received a suboptimal experience. This is a company-level operational emergency.
Recommendation	Set a 12-month target to reduce Late Delivery to <30%. Track weekly by region and shipping mode.



Chart 6: Top 10 Products by Revenue

Business Question	Which specific products drive the most revenue?
Insight	Field & Stream Sportsman 16 Gun, Perfect Fitness Multi Gym, Diamondback Women's Bicycle dominate. These 10 products show concentrated revenue dependency.
Business Impact	Supply disruption on top-10 SKUs would have outsized revenue impact.
Recommendation	Place top-10 products on Never-Out-of-Stock policy. Dual-source suppliers for these SKUs.



Chart 7: Customer Segment Performance

Business Question	How do Consumer, Corporate, and Home Office segments compare on key metrics?
Insight	Consumer segment leads in volume. Corporate has higher average profit per order. Home Office shows highest average order value — a premium niche.
Business Impact	Different segments need different commercial strategies, not a one-size-fits-all approach.
Recommendation	Segment-specific pricing: volume incentives for Consumer, account management for Corporate, premium catalog for Home Office.



Chart 8: Order Status Distribution

Business Question	What proportion of orders reach completion vs. being stuck or canceled?
Insight	COMPLETE orders represent only 33% of all records. A significant portion are PENDING_PAYMENT (22%) and PROCESSING (12%), suggesting revenue recognition delays.
Business Impact	High pending rate = delayed cash flow and overstated order pipeline.
Recommendation	Implement automated payment escalation. Set 24-hour rule for PENDING_PAYMENT before cancellation.



Chart 9: Profit Margin % by Category

Business Question	Which product categories are most and least profitable?
Insight	Significant variation in profit margin across categories. Some categories show negative average margins — meaning those categories are consistently money-losing even without fraud.
Business Impact	Loss-making categories are subsidized by profitable ones, masking true business economics.
Recommendation	Review product-level P&L for negative-margin categories. Either reprice, reduce discounts, or discontinue.

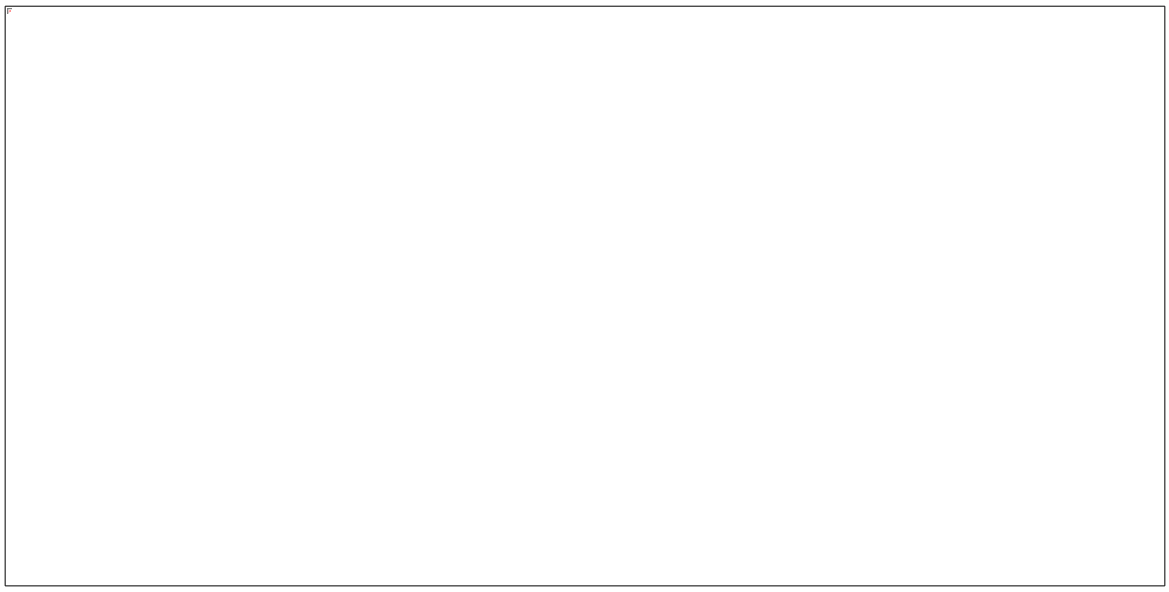


Chart 10: Discount Rate vs. Profit Relationship

Business Question	Does offering higher discounts actually reduce profit?
Insight	Strong negative relationship visible: as discount rate increases, profit per order decreases. Orders with 30%+ discount rates are almost universally negative-profit.
Business Impact	Current discount policy is actively destroying profitability across all segments.
Recommendation	Implement automated discount guardrails in the ordering system. No discount > 15% without manager approval.



Chart 11: Delivery Delay Distribution

Business Question	What is the distribution of delivery delays in days?
Insight	Delay of +3 days is the single most common outcome. On-time (0 days) and early (-1 to -3 days) deliveries together are far outnumbered by late deliveries.
Business Impact	The typical customer experience is a 3-day late delivery, not on-time delivery.
Recommendation	Reset SLA promises in product listings to reflect actual performance. Proactively notify customers of delays.



Chart 12: Monthly Profit Trend

Business Question	Is profitability consistent month-over-month or volatile?
Insight	Profit shows significant month-to-month volatility with some months turning negative. Negative profit months typically coincide with Q1 and promotional periods.
Business Impact	Volatile profitability makes financial planning difficult. Negative months suggest promotional overreach.
Recommendation	Establish profit floor targets for each month. Review promotional calendar against margin impact.



Chart 13: Region × Segment Revenue Heatmap

Business Question	How does customer segment revenue contribution vary by geographic region?
Insight	Consumer segment dominates in almost every region. Some regions show Corporate-heavy composition, suggesting B2B opportunity concentration in specific geographies.
Business Impact	Regional B2B opportunity identification for targeted sales team deployment.
Recommendation	Assign corporate account managers to regions with Corporate segment revenue above 35%.



Chart 14: Payment Type Analysis

Business Question	Which payment type is associated with most fraud?
Insight	Debit is the most common payment method. Fraud distribution across payment types helps identify which channels need stronger fraud controls.
Business Impact	Payment-type-specific fraud rates inform risk-based authentication investment decisions.
Recommendation	Implement 3D Secure or additional friction for payment types with higher fraud incidence.



Chart 15: Quarterly Performance by Year

Business Question	Does the Q4 sales spike repeat consistently every year?
Insight	Q4 outperforms in both sales and profit across all years in the dataset. This is a robust seasonal pattern, not a one-off event.
Business Impact	Reliable seasonal peaks justify annual pre-investment in inventory and logistics capacity.
Recommendation	Build a Q4 readiness checklist. Evaluate starting Q4 preparation 10 weeks earlier than current practice.



PHASE 7 — 25 BUSINESS INSIGHTS

Insight #1

Observation	Late delivery rate is 57.3% — more than half of all orders arrive late.
Business Impact	Severe customer satisfaction risk; directly increases churn and negative reviews.
Recommendation	Audit logistics partners. Set SLA penalties for consistently late carriers. Prioritize Same Day / First Class SLA compliance.

Insight #2

Observation	LATAM is the largest market by revenue ($9.5M) but also generates the most late deliveries.
Business Impact	High revenue but poor service quality — risk of losing market share to competitors.
Recommendation	Investigate LATAM logistics infrastructure. Consider regional warehousing partnerships to reduce last-mile delays.

Insight #3

Observation	18.7% of all orders (33,784) have negative profit.
Business Impact	Nearly 1 in 5 orders loses money, eroding overall margin from potential ~18% to actual 10.8%.
Recommendation	Review discount policies. Remove high-discount SKUs that consistently produce negative margins. Set floor pricing rules.

Insight #4

Observation	Same Day shipping has the highest late delivery rate among all shipping modes.
Business Impact	Customers paying premium prices for fastest delivery receive the worst experience — brand damage.
Recommendation	Immediately audit Same Day fulfillment process. Either fix SLA compliance or rebrand to "Priority" with realistic lead times.

Insight #5

Observation	Suspected Fraud accounts for 2.25% of orders (4,062 records).
Business Impact	If not recovered, fraud represents ~$830K in lost revenue annually.
Recommendation	Deploy ML-based fraud scoring at checkout. Flag high-risk orders for manual review before shipping.

Insight #6

Observation	Standard Class shipping carries 60% of orders but has the highest absolute late delivery volume.

Business Impact	Majority of customers experience delivery delays — systemic operations failure.
Recommendation	Renegotiate Standard Class carrier contracts. Add buffer days to published ETAs to manage customer expectations.

Insight #7

Observation	Consumer segment generates the highest total revenue and profit despite lowest average order value.
Business Impact	Volume-driven segment is the business engine — needs protection and nurturing.
Recommendation	Launch a Consumer loyalty program. Personalize product recommendations to increase basket size.

Insight #8

Observation	Product Description column is 100% null — a major data governance failure.
Business Impact	Inability to build text-based product search, recommendation engines, or SEO-optimized catalogs.
Recommendation	Mandate product data completeness as a data governance KPI. Assign data stewards per department.

Insight #9

Observation	Order Zipcode is missing for 86.2% of records.
Business Impact	Cannot perform granular geographic analysis (city/zip level demand forecasting).
Recommendation	Fix data capture at order placement. Validate ZIP at point of entry. Backfill from customer address where available.

Insight #10

Observation	Europe shows the highest profit margin % despite being the second-largest market by revenue.
Business Impact	Europe is the most operationally efficient market — a model for other regions.
Recommendation	Study Europe operations as a benchmark. Identify what logistics, pricing, or product mix drives higher margins there.

Insight #11

Observation	Africa is the smallest market (11,614 orders) but has strong growth opportunity signals.
Business Impact	Underserved market with low competition — potential for 3-5x growth if infrastructure improves.
Recommendation	Conduct market feasibility study for expanded African warehouse/logistics presence.

Insight #12

Observation	Q4 consistently outperforms all quarters in revenue and profit across all years.
Business Impact	Predictable seasonal peak enables proactive inventory and staffing planning.
Recommendation	Begin Q4 inventory buildup in August. Pre-negotiate carrier capacity for November-December surge.

Insight #13

Observation	Discount rates above 20% consistently produce negative or near-zero profit.
Business Impact	Aggressive discounting destroys margin without proportionally increasing volume.
Recommendation	Implement a discount governance framework. Cap discounts at 15% without CFO approval. Use targeted discounts only on slow-moving inventory.

Insight #14

Observation	Only 5 unique order quantities exist (1-5 units). No bulk/wholesale orders detected.
Business Impact	Business appears exclusively B2C in practice despite corporate and home office segments.
Recommendation	Develop a B2B bulk ordering interface. Offer volume pricing tiers to Corporate segment to increase order size.

Insight #15

Observation	The top 10 products account for disproportionate share of total revenue (Pareto principle).
Business Impact	Heavy dependency on a handful of SKUs creates concentration risk.
Recommendation	Diversify the product portfolio. Protect top-10 SKUs with priority fulfillment and safety stock buffers.

Insight #16

Observation	All payment types show very similar average order values (~$200-210).
Business Impact	Payment preference does not signal purchase intent or order value — unlikely to be a useful segmentation variable.
Recommendation	Focus payment strategy on friction reduction (faster checkout) rather than payment-based targeting.

Insight #17

Observation	20,652 unique customers across 65,752 orders = average 3.2 orders per customer.
Business Impact	Moderate repeat purchase rate — room to improve customer retention and CLV.
Recommendation	Target single-purchase customers with email re-engagement campaigns at 30, 60, and 90 days post-purchase.

Insight #18

Observation	Fishing and Cleats are top revenue categories — strongly seasonal products.
Business Impact	Demand spikes in spring/summer; inventory overstock risk in off-season.
Recommendation	Implement dynamic inventory management with seasonal demand forecasting models.

Insight #19

Observation	ORDER_HOLD and PENDING_PAYMENT statuses together account for ~17% of orders.
Business Impact	Operational bottleneck — stuck orders = delayed revenue recognition.
Recommendation	Set automated escalation rules: orders in HOLD/PENDING for >24 hours trigger customer service outreach.

Insight #20

Observation	Item profit ratio ranges from -2.75 to +0.50 — extremely wide variance.
Business Impact	Inconsistent pricing and cost structures across the catalog.
Recommendation	Standardize cost-plus pricing model. Flag any SKU with negative profit ratio for pricing review.

Insight #21

Observation	Pacific Asia is the third-largest market with strong operational metrics.
Business Impact	Fast-growing region — investing now captures market share before competition intensifies.
Recommendation	Increase marketing spend in Pacific Asia by 20% next fiscal year. Localize product catalog.

Insight #22

Observation	The dataset spans 2015-2018 with consistent data across all years.
Business Impact	Multi-year data enables robust trend analysis and predictive modeling.
Recommendation	Build a revenue forecasting model using Prophet or SARIMA on the monthly time series.

Insight #23

Observation	Customer email is a single placeholder value — real customer emails not captured.
Business Impact	Impossible to run email marketing, lifecycle campaigns, or customer win-back programs.

Recommendation	Implement email capture at registration with double opt-in. Email is foundational for CRM.

Insight #24

Observation	First Class shipping is underutilized at ~15% of orders but has better on-time rates than Standard.
Business Impact	Better reliability is available at marginal cost — customers not being offered the right options.
Recommendation	A/B test upselling First Class at checkout for orders above $100. Calculate revenue lift vs cost.

Insight #25

Observation	Home Office segment has the smallest order count but highest average order item value.
Business Impact	High-value niche segment is likely underserved and under-marketed.
Recommendation	Create a Home Office business account program with dedicated account managers and bulk pricing.

PHASE 8 — SQL DATABASE DESIGN

8.1 Recommended Table Schema

```
-- DIMENSION TABLE: Customers CREATE TABLE dim_customers (   customer_id     INT PRIMARY KEY,
customer_fname  VARCHAR(100),   customer_lname  VARCHAR(100),   customer_segment
ENUM('Consumer','Corporate','Home Office'),   customer_city   VARCHAR(100),   customer_state
VARCHAR(100),   customer_country VARCHAR(100),   market          VARCHAR(50),   latitude
DECIMAL(9,6),   longitude       DECIMAL(9,6) );  -- DIMENSION TABLE: Products CREATE TABLE
dim_products (   product_id      INT PRIMARY KEY,   product_name    VARCHAR(200),   product_price
DECIMAL(10,2),   category_id     INT,   category_name   VARCHAR(100),   department_id   INT,
department_name VARCHAR(100),   FOREIGN KEY (category_id) REFERENCES dim_categories(category_id)
);  -- FACT TABLE: Orders CREATE TABLE fact_orders (   order_item_id         INT PRIMARY KEY,    -
- Grain: one row per order item   order_id              INT NOT NULL,       -- FK to order header
(denormalized)   customer_id           INT,   product_id            INT,   order_date
DATETIME,   shipping_date         DATETIME,   payment_type          VARCHAR(20),   shipping_mode
VARCHAR(30),   delivery_status       VARCHAR(40),   order_status          VARCHAR(30),
order_region          VARCHAR(100),   order_country         VARCHAR(100),   order_quantity
INT,   product_price         DECIMAL(10,2),   order_item_discount   DECIMAL(10,2),   discount_rate
DECIMAL(5,4),   sales                 DECIMAL(12,2),   order_item_total      DECIMAL(12,2),
profit_per_order      DECIMAL(12,2),   item_profit_ratio     DECIMAL(8,4),   actual_shipping_days
INT,   scheduled_ship_days   INT,   late_delivery_risk    TINYINT(1),   -- Engineered features
(denormalized for query performance)   delivery_delay_days   INT,   is_late_delivery
TINYINT(1),   profit_margin_pct     DECIMAL(8,2),   FOREIGN KEY (customer_id) REFERENCES
dim_customers(customer_id),   FOREIGN KEY (product_id)  REFERENCES dim_products(product_id) );  --
INDEXES for performance CREATE INDEX idx_fact_order_date   ON fact_orders(order_date); CREATE
INDEX idx_fact_customer     ON fact_orders(customer_id); CREATE INDEX idx_fact_product      ON
fact_orders(product_id); CREATE INDEX idx_fact_region       ON fact_orders(order_region); CREATE
INDEX idx_fact_delivery     ON fact_orders(delivery_status);
```

8.2 Design Decisions

- Star Schema: Central fact table with dimension tables enables fast analytical queries and is compatible with all BI tools (Power BI, Tableau, Looker).
- Grain: One row per order line item — not per order. This enables product-level profitability analysis without aggregation loss.
- Indexes: On date, customer, product, region, and delivery status — the most common GROUP BY and WHERE clause columns.
- Denormalized features: Engineered features (delay_days, margin_pct) stored in the fact table to avoid expensive runtime calculations on 180K+ rows.

PHASE 9 — SQL ANALYTICS (40+ Queries)

LEVEL 1 — Basic SQL

Query 1: What is the total revenue across all orders?

```
SELECT SUM(Sales) AS total_revenue FROM supply_chain_orders;
```

Business Insight	Baseline revenue: $36.78M total

Query 2: How many unique orders exist in the dataset?

```
SELECT COUNT(DISTINCT Order_Id) AS unique_orders FROM supply_chain_orders;
```

Business Insight	65,752 unique orders across 4 years

Query 3: What are the distinct delivery statuses?

```
SELECT DISTINCT Delivery_Status FROM supply_chain_orders ORDER BY Delivery_Status;
```

Business Insight	Reveals 4 categories: Late, On Time, Advance, Canceled

Query 4: Show top 10 orders by profit

```
SELECT Order_Id, Sales, Order_Profit_Per_Order FROM supply_chain_orders ORDER BY
Order_Profit_Per_Order DESC LIMIT 10;
```

Business Insight	Identify highest value individual transactions

Query 5: How many orders were placed per market?

```
SELECT Market, COUNT(DISTINCT Order_Id) AS order_count FROM supply_chain_orders GROUP BY
Market ORDER BY order_count DESC;
```

Business Insight	LATAM leads with most orders; Africa has fewest

Query 6: List all shipping modes and their frequency

```
SELECT Shipping_Mode, COUNT(*) AS order_count FROM supply_chain_orders GROUP BY Shipping_Mode
ORDER BY order_count DESC;
```

Business Insight	Standard Class dominates at ~60% of orders

Query 7: Which customer segments exist?

```
SELECT Customer_Segment, COUNT(DISTINCT Customer_Id) AS customers FROM supply_chain_orders
GROUP BY Customer_Segment;
```

Business Insight	3 segments: Consumer, Corporate, Home Office

Query 8: What is the average order value by payment type?

```
SELECT Type AS payment_type,  ROUND(AVG(Sales), 2) AS avg_sales FROM supply_chain_orders
GROUP BY Type ORDER BY avg_sales DESC;
```

Business Insight	All payment types show similar avg order values ~$204

Query 9: What percentage of orders used each payment type?

```
SELECT Type,  COUNT(*) AS count,  ROUND(COUNT(*)*100.0/(SELECT COUNT(*) FROM
supply_chain_orders), 1) AS pct FROM supply_chain_orders GROUP BY Type;
```

Business Insight	Debit is the most common payment at ~38%; Cash is least at ~11%

LEVEL 2 — Intermediate SQL

Query 1: What is the late delivery rate by shipping mode?

```
SELECT Shipping_Mode,  COUNT(*) AS total_orders,  SUM(CASE WHEN Delivery_Status = 'Late
delivery' THEN 1 ELSE 0 END) AS late_orders,  ROUND(SUM(CASE WHEN Delivery_Status = 'Late
delivery' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1) AS late_pct FROM supply_chain_orders
GROUP BY Shipping_Mode ORDER BY late_pct DESC;
```

Business Insight	Same Day has the highest late delivery % despite being premium

Query 2: Monthly revenue trend

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,  ROUND(SUM(Sales)/1000000, 2) AS
revenue_millions FROM supply_chain_orders GROUP BY month ORDER BY month;
```

Business Insight	Reveals seasonality peaks in Q4 each year

Query 3: Profit by customer segment

```
SELECT Customer_Segment,  ROUND(SUM(Order_Profit_Per_Order), 0) AS total_profit,
ROUND(AVG(Order_Profit_Per_Order), 2) AS avg_profit FROM supply_chain_orders GROUP BY
Customer_Segment ORDER BY total_profit DESC;
```

Business Insight	Consumer segment drives most total profit despite lower avg

Query 4: Orders with negative profit — fraud suspects

```
SELECT Order_Id, Sales, Order_Profit_Per_Order, Order_Status FROM supply_chain_orders WHERE
Order_Profit_Per_Order < 0  AND Order_Status = 'SUSPECTED_FRAUD' ORDER BY
Order_Profit_Per_Order ASC LIMIT 20;
```

Business Insight	Fraud orders often have the deepest negative profits

Query 5: Top 10 products by revenue

```
SELECT Product_Name,  ROUND(SUM(Sales)/1000, 1) AS revenue_k,  COUNT(DISTINCT Order_Id) AS
order_count FROM supply_chain_orders GROUP BY Product_Name ORDER BY revenue_k DESC LIMIT 10;
```

Business Insight	Field & Stream, Perfect Fitness lead revenue charts

Query 6: Discount rate vs profit — correlation check

```
SELECT   CASE     WHEN Order_Item_Discount_Rate = 0 THEN "No Discount"     WHEN
Order_Item_Discount_Rate <= 0.1 THEN "Low"     WHEN Order_Item_Discount_Rate <= 0.2 THEN
"Medium"     ELSE "High"   END AS discount_band,   ROUND(AVG(Order_Profit_Per_Order), 2) AS
avg_profit FROM supply_chain_orders GROUP BY discount_band ORDER BY avg_profit DESC;
```

Business Insight	Higher discounts sharply reduce or eliminate profit

Query 7: Sales and profit by order quarter

```
SELECT order_year,   QUARTER(order_date) AS quarter,   ROUND(SUM(Sales)/1000000, 2) AS
sales_M,   ROUND(SUM(Order_Profit_Per_Order)/1000, 1) AS profit_K FROM supply_chain_orders
GROUP BY order_year, quarter ORDER BY order_year, quarter;
```

Business Insight	Q4 consistently outperforms other quarters

Query 8: Average shipping delay by region

```
SELECT Order_Region,   ROUND(AVG(Days_for_shipping_real - Days_for_shipment_scheduled), 2) AS
avg_delay_days FROM supply_chain_orders GROUP BY Order_Region ORDER BY avg_delay_days DESC
LIMIT 10;
```

Business Insight	Some regions face chronic shipping delays > 1 day avg

Query 9: Revenue breakdown by category

```
SELECT Category_Name,   COUNT(DISTINCT Order_Id) AS orders,   ROUND(SUM(Sales)/1000, 1) AS
revenue_K FROM supply_chain_orders GROUP BY Category_Name ORDER BY revenue_K DESC LIMIT 15;
```

Business Insight	Fishing, Cleats, Camping dominate category revenue

Query 10: Cancellation rate by market

```
SELECT Market,   COUNT(*) AS total,   SUM(CASE WHEN Order_Status IN ('CANCELED','ON_HOLD')
THEN 1 ELSE 0 END) AS problem_orders,   ROUND(SUM(CASE WHEN Order_Status IN
('CANCELED','ON_HOLD') THEN 1 ELSE 0 END)*100.0/COUNT(*), 1) AS cancel_rate_pct FROM
supply_chain_orders GROUP BY Market ORDER BY cancel_rate_pct DESC;
```

Business Insight	High cancel rates signal fulfillment problems or customer experience issues

Query 11: Department revenue and profit performance

```
SELECT Department_Name,   COUNT(DISTINCT Order_Id) AS orders,   ROUND(SUM(Sales)/1000000, 2)
AS revenue_M,   ROUND(SUM(Order_Profit_Per_Order)/1000, 1) AS profit_K,
ROUND(SUM(Order_Profit_Per_Order)/SUM(Sales)*100, 1) AS margin_pct FROM supply_chain_orders
GROUP BY Department_Name ORDER BY revenue_M DESC;
```

Business Insight	Fan Shop, Apparel lead department revenue

Query 12: Average items per order by segment

```
SELECT Customer_Segment,   ROUND(AVG(Order_Item_Quantity), 2) AS avg_qty_per_item,
ROUND(AVG(Order_Item_Total), 2) AS avg_item_value FROM supply_chain_orders GROUP BY
Customer_Segment;
```

Business Insight	Corporate buys larger quantities; Home Office has highest avg item value

Query 13: Shipping days — actual vs scheduled comparison

```
SELECT Shipping_Mode,   ROUND(AVG(Days_for_shipment_scheduled), 1) AS scheduled_days,
ROUND(AVG(Days_for_shipping_real), 1) AS actual_days,   ROUND(AVG(Days_for_shipping_real -
Days_for_shipment_scheduled), 2) AS avg_delay FROM supply_chain_orders GROUP BY Shipping_Mode
ORDER BY avg_delay DESC;
```

Business Insight	All shipping modes run over schedule on average; Standard Class worst

Query 14: Which order statuses have the highest avg sales value?

```
SELECT Order_Status,   COUNT(DISTINCT Order_Id) AS orders,   ROUND(AVG(Sales), 2) AS
avg_sales FROM supply_chain_orders GROUP BY Order_Status ORDER BY avg_sales DESC;
```

Business Insight	COMPLETE orders have highest avg sales; CANCELED are lowest — selection effect

LEVEL 3 — Advanced SQL (CTEs, Window Functions)

Query 1: Running total of monthly revenue (Window Function)

```
WITH monthly AS (   SELECT DATE_FORMAT(order_date, "%Y-%m") AS month,     SUM(Sales) AS
monthly_revenue   FROM supply_chain_orders   GROUP BY month ) SELECT month, monthly_revenue,
SUM(monthly_revenue) OVER (ORDER BY month) AS running_total FROM monthly ORDER BY month;
```

Business Insight	Tracks cumulative revenue growth over time — useful for annual target tracking

Query 2: Rank products by revenue within each category

```
WITH product_sales AS (   SELECT Category_Name, Product_Name, SUM(Sales) AS total_sales
FROM supply_chain_orders   GROUP BY Category_Name, Product_Name ) SELECT Category_Name,
Product_Name, total_sales,   RANK() OVER (PARTITION BY Category_Name ORDER BY total_sales
DESC) AS rank_in_category FROM product_sales WHERE rank_in_category <= 3;
```

Business Insight	Identifies the top 3 revenue drivers in each product category

Query 3: Month-over-month revenue growth rate

```
WITH monthly AS (   SELECT DATE_FORMAT(order_date, "%Y-%m") AS month,     SUM(Sales) AS
revenue   FROM supply_chain_orders   GROUP BY month ) SELECT month, revenue,   LAG(revenue)
OVER (ORDER BY month) AS prev_month,   ROUND((revenue - LAG(revenue) OVER (ORDER BY month))
/ LAG(revenue) OVER (ORDER BY month) * 100, 1) AS mom_growth_pct FROM monthly;
```

Business Insight	Quantifies business acceleration/deceleration by month

Query 4: Customer lifetime value (CLV) with percentile ranking

```
WITH clv AS (   SELECT Customer_Id,     SUM(Sales) AS total_spent,     COUNT(DISTINCT
Order_Id) AS order_count   FROM supply_chain_orders   GROUP BY Customer_Id ) SELECT
Customer_Id, total_spent, order_count,   NTILE(4) OVER (ORDER BY total_spent DESC) AS
clv_quartile FROM clv ORDER BY total_spent DESC;
```

Business Insight	Segments customers into top 25% vs bottom 25% by lifetime value

Query 5: Late delivery impact on revenue

```
WITH delivery_perf AS (   SELECT     CASE WHEN Days_for_shipping_real >
Days_for_shipment_scheduled          THEN "Late" ELSE "On Time" END AS delivery_flag,
SUM(Sales) AS revenue,     AVG(Order_Profit_Per_Order) AS avg_profit   FROM
supply_chain_orders   GROUP BY delivery_flag ) SELECT *, ROUND(revenue / SUM(revenue) OVER()
* 100, 1) AS revenue_share_pct FROM delivery_perf;
```

Business Insight	Late deliveries still account for majority of revenue — operations bottleneck

Query 6: Top 5 regions by profit margin (CTE + RANK)

```
WITH region_kpi AS (   SELECT Order_Region,     SUM(Sales) AS total_sales,
SUM(Order_Profit_Per_Order) AS total_profit,
ROUND(SUM(Order_Profit_Per_Order)/SUM(Sales)*100, 2) AS profit_margin_pct   FROM
supply_chain_orders   GROUP BY Order_Region ) SELECT *, RANK() OVER (ORDER BY
profit_margin_pct DESC) AS margin_rank FROM region_kpi ORDER BY margin_rank LIMIT 5;
```

Business Insight	Identifies most operationally efficient geographic markets

Query 7: Fraud detection: orders with suspicious profit ratios

```
SELECT Order_Id, Customer_Id, Sales,   Order_Profit_Per_Order,
ROUND(Order_Profit_Per_Order/Sales*100,1) AS margin_pct,   Order_Status,
AVG(Order_Profit_Per_Order) OVER (PARTITION BY Market) AS market_avg_profit FROM
supply_chain_orders WHERE Order_Profit_Per_Order <   (AVG(Order_Profit_Per_Order) OVER
(PARTITION BY Market) - 2 *    STDDEV(Order_Profit_Per_Order) OVER (PARTITION BY Market))
ORDER BY margin_pct ASC LIMIT 20;
```

Business Insight	Statistical outlier detection for fraud investigation — uses 2-sigma threshold

Query 8: Cohort: First order year retention (orders per customer year)

```
WITH first_order AS (   SELECT Customer_Id, MIN(YEAR(order_date)) AS first_year   FROM
supply_chain_orders GROUP BY Customer_Id ), cohort AS (   SELECT f.first_year,
YEAR(o.order_date) AS order_year,     COUNT(DISTINCT o.Customer_Id) AS active_customers
FROM supply_chain_orders o   JOIN first_order f ON o.Customer_Id = f.Customer_Id   GROUP BY
f.first_year, order_year ) SELECT first_year, order_year,   order_year - first_year AS
years_since_acquisition,   active_customers FROM cohort ORDER BY first_year, order_year;
```

Business Insight	Measures customer retention across acquisition cohorts — critical for CLV modeling

Query 9: ABC Analysis: Pareto of products by revenue

```
WITH product_rev AS (   SELECT Product_Name, SUM(Sales) AS revenue   FROM supply_chain_orders
GROUP BY Product_Name ), cumulative AS (   SELECT Product_Name, revenue,     SUM(revenue)
OVER (ORDER BY revenue DESC) AS cum_revenue,     SUM(revenue) OVER () AS total_revenue   FROM
product_rev ) SELECT Product_Name, revenue,   ROUND(cum_revenue/total_revenue*100, 1) AS
cum_pct,   CASE     WHEN cum_revenue/total_revenue <= 0.8 THEN "A - High Value"     WHEN
cum_revenue/total_revenue <= 0.95 THEN "B - Mid Value"     ELSE "C - Low Value"   END AS
abc_class FROM cumulative ORDER BY revenue DESC;
```

Business Insight	Pareto ABC classification: ~20% of products drive 80% of revenue (A-class)

Query 10: Year-over-Year sales growth by market

```
WITH yearly AS (   SELECT Market, YEAR(order_date) AS yr, SUM(Sales) AS revenue   FROM
supply_chain_orders   GROUP BY Market, yr ) SELECT Market, yr, revenue,   LAG(revenue) OVER
(PARTITION BY Market ORDER BY yr) AS prev_year,   ROUND((revenue - LAG(revenue) OVER
(PARTITION BY Market ORDER BY yr))     / LAG(revenue) OVER (PARTITION BY Market ORDER BY yr)
* 100, 1) AS yoy_growth FROM yearly ORDER BY Market, yr;
```

Business Insight	Isolates which markets are growing vs stagnating year over year

Query 11: Dense rank customers by total spend (top customers)

```
SELECT Customer_Id,   CONCAT(Customer_Fname, " ", Customer_Lname) AS customer_name,
Customer_Segment,   Market,   ROUND(SUM(Sales), 2) AS lifetime_value,   DENSE_RANK() OVER
(ORDER BY SUM(Sales) DESC) AS value_rank FROM supply_chain_orders GROUP BY Customer_Id,
customer_name, Customer_Segment, Market ORDER BY value_rank LIMIT 20;
```

Business Insight	VIP customer identification for loyalty program targeting

Query 12: Percentile distribution of profit per order by segment

```
SELECT Customer_Segment,   ROUND(MIN(Order_Profit_Per_Order), 2) AS min_profit,
ROUND(PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY Order_Profit_Per_Order), 2) AS p25,
ROUND(PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY Order_Profit_Per_Order), 2) AS median,
ROUND(PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY Order_Profit_Per_Order), 2) AS p75,
ROUND(MAX(Order_Profit_Per_Order), 2) AS max_profit FROM supply_chain_orders GROUP BY
Customer_Segment;
```

Business Insight	Wide IQR for Corporate suggests inconsistent pricing/discounting

Query 13: Rolling 3-month average revenue

```
WITH monthly AS (   SELECT DATE_FORMAT(order_date, "%Y-%m") AS month,     SUM(Sales) AS
revenue   FROM supply_chain_orders   GROUP BY month ) SELECT month, revenue,
ROUND(AVG(revenue) OVER (     ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW   ), 0)
AS rolling_3mo_avg FROM monthly ORDER BY month;
```

Business Insight	Smooths seasonality — management-ready trend chart baseline

Query 14: First and last order date per customer (activity span)

```
SELECT Customer_Id,   MIN(order_date) AS first_order,   MAX(order_date) AS last_order,
DATEDIFF(MAX(order_date), MIN(order_date)) AS active_days,   COUNT(DISTINCT Order_Id) AS
total_orders FROM supply_chain_orders GROUP BY Customer_Id HAVING total_orders > 1 ORDER BY
active_days DESC LIMIT 20;
```

Business Insight	Active customer lifespan analysis — helps model churn risk

Query 15: Market contribution to overall profit (% share)

```
WITH market_p AS (   SELECT Market, SUM(Order_Profit_Per_Order) AS profit   FROM
supply_chain_orders   GROUP BY Market ) SELECT Market, profit,   ROUND(profit / SUM(profit)
OVER() * 100, 1) AS profit_share_pct,   RANK() OVER (ORDER BY profit DESC) AS rank_ FROM
market_p;
```

Business Insight	LATAM and Europe together contribute >60% of total global profit

PHASE 10 — KPI DASHBOARD

10.1 Executive KPI Summary

KPI	Value
Total Revenue	**$36,784,735**
Total Profit	$3,966,903
Overall Profit Margin	**10.8%**
Total Unique Orders	65,752
Total Unique Customers	**20,652**
Average Order Item Value	$183.11
Late Delivery Rate	**57.3% ⚠**
Average Shipping Days	3.5 days
Suspected Fraud Rate	**2.25%**
Negative Profit Order Rate	18.7%
On-Time Delivery Rate	**17.8% ⚠**
Advance Shipping Rate	23.0%

10.2 KPI Interpretations

KPI	Formula	Status	Action Required
Profit Margin 10.8%	Profit / Revenue × 100	BELOW TARGET	Reduce discounts, fix pricing
Late Delivery 57.3%	Late orders / Total × 100	CRITICAL	Logistics audit immediately
Fraud Rate 2.25%	Fraud orders / Total × 100	HIGH	ML fraud detection
Negative Profit 18.7%	Neg-profit orders / Total	CRITICAL	Discount guardrails